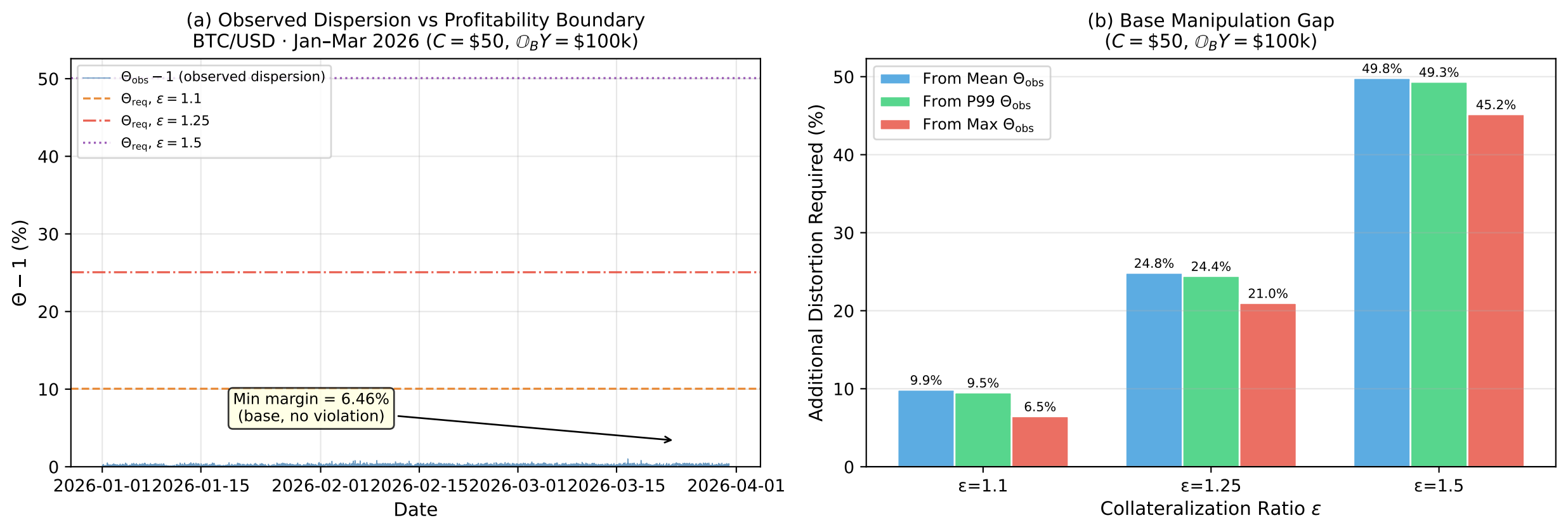

(a) Observed Dispersion vs Profitability Boundary
BTC/USD · Jan–Mar 2026 ($C = \$50$, $\mathcal{O}_B Y = \$100$k)
$\Theta_{obs} - 1$ (observed dispersion)
Θ_{req}, $\varepsilon = 1.1$
Θ_{req}, $\varepsilon = 1.25$
Θ_{req}, $\varepsilon = 1.5$
Min margin = 6.46%
(base, no violation)
$\Theta - 1$ (%)
Date
2026-01-01
2026-01-15
2026-02-01
2026-02-15
2026-03-01
2026-03-15
2026-04-01
(b) Base Manipulation Gap
($C = \$50$, $\mathcal{O}_B Y = \$100$k)
From Mean Θ_{obs}
From P99 Θ_{obs}
From Max Θ_{obs}
Additional Distortion Required (%)
9.9%
9.5%
6.5%
24.8%
24.4%
21.0%
49.8%
49.3%
45.2%
ε=1.1
ε=1.25
ε=1.5
Collateralization Ratio ε